

CACHE TRANSLATION ACROSS MODEL SCALE: AN AUDIT WITH IDENTITY CONTROLS AND ORACLE PROBES

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ABSTRACT

Cache translation maps the key-value (KV) cache that a large teacher model forms over a context into the cache space of a smaller student model, so that the student answers without re-prefilling the context. A growing line of work reports quality-preserving translation across heterogeneous models. We audit this claim on Qwen3 pairs (8B→1.7B, 8B→0.6B, 4B→1.7B, and 4B→0.6B), plus five cross-architecture students, with a three-hypothesis framework and two instruments that prior evaluations omit. First, an *identity control* injects the student’s own cache through the full translation-and-injection pipeline: it reproduces the student’s own prefill exactly (per-sample logit cosine 1.000), which isolates mechanics from mapping. Second, *oracle probes* mix teacher-cache content into the student’s own cache and probe what the student can extract from translated content. Across seven mapper families, a controlled calibration ladder on a fixed evaluation set, and 96 recorded audit-protocol runs, no configuration makes the strong student exceed its own prefill: gold-probability change spans -0.14 to -0.39 against a self-kv control at $+0.000$. A held-out reconstruction quantifies the structural cause: the best per-head mapper recovers student key states ($R^2 = +0.81$) but only partially recovers value states ($R^2 = +0.32$; per-(layer,head) mean over held-out calibration contexts; Section 6). Calibration budget is inert (CHG -0.249 at $c=30$ vs -0.244 at $c=200$ on the same evaluation set), and the tested nonlinear MLP mapper does not close the gap. The weak student’s point estimate is near parity ($+0.010$, CI $[-0.08, +0.10]$), but the interval is too wide to establish non-inferiority at the pre-specified margin. Oracle probes show a monotone decline as teacher content replaces student content in early and mid layers, robust to translator quality, and a perplexity–accuracy decoupling: near-native fluency (PPL 23.7 versus 21.2) still leaves accuracy at 0.267 versus 0.500. Under zero re-prefill, no configuration we test lets the student extract the teacher’s answer-relevant advantage from its cache; the evidence is consistent with an advantage that is weight-mediated rather than recoverable from raw KV states by the tested translators. The audit protocol doubles as a checklist that a future cache-translation claim can run in one evaluation.

1 INTRODUCTION

Multi-model serving systems want to reuse the teacher’s work. A large teacher prefills a long context once; a smaller student should inherit that state and answer at teacher quality without re-reading the context. Cache translation formalizes this idea: learn a map $\mathbf{KV}^T \mapsto \widehat{\mathbf{KV}}^S$ from the teacher’s cache space into the student’s. Recent systems report encouraging results, most prominently MoT, which preserves 51.0% closed-set QA accuracy at 7B→0.5B scale (Lee et al., 2026).

These reports conflate three questions that can fail independently: *mechanics* (does the pipeline consume a foreign cache correctly?), *mapping* (does the map replace the student’s own cache?), and *exploitability* (does the teacher’s cache carry student-readable content that lifts the student above its own prefill?). Prior evaluations score the translated end-to-end pipeline against the student, so a positive delta could come from any of the three; a null result cannot be attributed. The line also lacks an identity control, a probe of the exploitability ceiling, and any report of what happens to *accuracy* when perplexity is repaired.

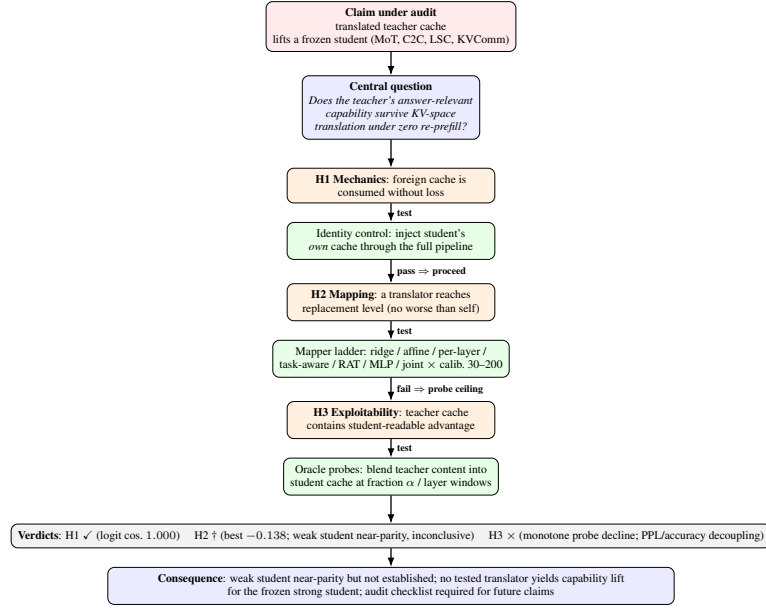


Figure 1: The three-hypothesis audit as a decision pipeline. Each hypothesis carries a dedicated instrument and a gate; H2’s failure routes the audit to H3’s oracle probes, which test whether teacher content is exploitable by the frozen student without training a translator.

We evaluate cache translation as an audit with three gated hypotheses. **H1 (mechanics)** asks whether a foreign cache is consumed without loss (instrument: an identity control that injects the student’s own cache through the full pipeline). **H2 (mapping)** asks whether any translator reaches replacement level (instrument: seven mapper families under a disjoint calibration ladder of 30 to 200 examples, including a residual-anchored translator, RAT, derived from the models’ own projections and a shared vocabulary alignment). **H3 (exploitability)** asks whether the teacher’s cache carries student-readable advantage at all (instrument: an oracle probe that mixes teacher-cache content into the student’s own cache at controlled fractions and layer windows; Figure 1 shows the gates and instruments).

The audit returns three verdicts on Qwen3 pairs. H1 passes: the identity control reproduces the student’s own prefill exactly (per-sample logit cosine 1.000), so mechanics is not the failure mode. H2 fails for the strong student: the gold-probability change over its own prefill spans -0.14 to -0.39 across the mapper ladder (Section 5), while the weak 0.6B student’s point estimate is near parity ($+0.010$, CI $[-0.081, +0.102]$), too imprecise to establish non-inferiority at the pre-specified margin. H3 returns a principled negative: with 75% student cache and only 25% teacher content, accuracy already drops from 0.500 to 0.367; the decline is monotone, and replacing the top third of layers is the only harmless configuration. A translator that restores near-native fluency (PPL 23.7 versus 21.2) still leaves accuracy at 0.267: a cache can read fluently and still miss the answer; the deficit is not a fluency problem.

Three things fall out of the audit. The first is a framework: identity controls, disjoint calibration/evaluation splits, gold-probability metrics with bootstrap intervals and permutation tests, and diagnostic oracle probes that measure whether teacher-cache content is exploitable without training a translator (Section 4). The second is a bounded design space: seven mapper families, a calibration ladder, and component ablations under one protocol; we additionally derive the most architecture-informed translator we could (RAT), and it fails too, which strengthens the negative. The third is a verdict with consequences: mechanics passes, replacement fails for the strong student and is inconclusive for the weak one, no probe configuration with teacher content beats the student’s own cache, and perplexity cannot stand in for accuracy as evidence (Section 5, Section 7). H2’s failure routes the audit to H3 because the probes measure what a frozen student can extract from teacher-cache content without training a translator, so a null result cannot be blamed on the mapper.

2 A SURVEY OF CACHE-LEVEL COMMUNICATION BETWEEN MODELS

We organize prior work into four lines, each closing with the assumption it leaves untested; Table 1 summarizes the comparison.

2.1 KV REUSE WITHIN A SINGLE MODEL

Prompt caching, cache-augmented generation, and streaming attention reuse a model’s own cache across requests: attention sinks (Xiao et al., 2023), cache compression and streaming (Liu et al., 2024), selective recomputation (Yao et al., 2025), and layer-wise cache sharing (Yang et al., 2024). These systems assume a single model on both sides of the cache, so the reuse question is about storage and attention cost, not representation compatibility. Our audit inherits their serving motivation but moves the mismatch to the model axis.

2.2 CROSS-MODEL CACHE TRANSLATION

The line closest to our audit translates caches between models: C2C via depth-ratio layer mapping (Fu et al., 2025), LSC through a shared latent space with a cross-attention translator (Dery et al., 2026), MoT through gated translator modules with a Context Correction Loss that aligns a replayed target trajectory with the native one—reporting 51.0% average closed-set QA accuracy at Qwen2.5-7B→0.5B and 96.3% retention in long-context cache-augmented generation (Lee et al., 2026)—and KVComm through attention-ranked KV layers (Shi et al., 2025); Interlat communicates last-hidden-state messages (Du et al., 2025), and HCache restores caches from hidden-state translations (Gao et al., 2025). Two properties of this line matter for our audit. First, its evaluations score the translated pipeline against the student baseline, which conflates mechanics, mapping, and exploitability. Second, its strongest variants do not operate under strict zero re-prefill: MoT reconstructs the target-side cache by replaying the context through the target model with source-guided sparse attention, so the reported quality includes a correction pass that a cold-start deployment would not get. A naive target-side replay pass does not by itself recover the strong student (Appendix E), so MoT’s gains cannot be attributed to replay alone. Our audit holds the zero-re-prefill constraint fixed and asks what a translator alone can deliver. Our numbers sit far below the 73–98% accuracy retention that Heo et al. (Heo et al., 2026) report for a closed-form ridge mapper with top- k multi-source-layer selection and RoPE-stripped key mapping (four of six matched-KV pairs in three families). Three differences separate the claims. First, the metrics differ: they report accuracy *retained relative to the receiver’s standalone prefill*; we report the gold-probability change over the student’s own prefill under disjoint splits, so a high retention score is compatible with a small systematic deficit. Second, their top- k cross-layer selection and de-RoPE key map are not in our ladder, so our negative result is scoped to the translator families we test. Third, their pairs sit at large scale (e.g. Qwen3-14B→32B) with a 500×1024 -token pool; ours span 8B→1.7B with 30–200 examples. Under the strict regime we measure gold-probability changes of -0.25 to -0.30 ; the gap persists across our tested translator families and protocol, and we do not claim it extends to their full design space. Heo et al. (Heo et al., 2026) also report that top- k source layers explain 56–79% of target key variance—a positive signal that the per-head underdetermination we measure (Section 6) is regime-dependent, and a concrete target for the cross-layer, de-RoPE design space our audit does not cover.

2.3 CROSS-MODEL REPRESENTATION ALIGNMENT

Model stitching shows representation quality varying sharply across runs and layers (Bansal et al., 2021); relative representations enable zero-shot encoder communication (Moschella et al., 2022); the Platonic Representation Hypothesis argues representations converge up to a linear transform (Huh et al., 2024), with refinements and proofs (Gröger et al., 2026; Ziyin & Chuang, 2025); and vec2vec translates embeddings without paired data (Jha et al., 2025). These results motivate the hypothesis that a linear map between cache spaces should exist, but the convergence evidence comes from pooled embedding spaces, while a cache is a per-layer, per-head, position-indexed state whose student-relevant content is not determined by its teacher counterpart (Section 6).

Table 1: Cache-translation and adjacent work, organized by what is translated and what is held fixed. “Replay” marks a target-side correction pass; our audit holds strict zero re-prefill. No cited evaluation isolates mechanics with an identity injection or reports accuracy at matched perplexity; MoT’s injection-window ablations are the closest antecedent (Section 2).

Work	What moves	Alignment	Replay	Unexamined assumption
C2C (Fu et al., 2025)	Full KV	depth-ratio layers	no	a linear map per channel suffices
LSC (Dery et al., 2026)	KV via shared latent	learned latent space	no	one latent space fits all layers
MoT (Lee et al., 2026)	KV windows	gated translators	yes (sparse)	pipeline delta $\neq$ translation value
Heo et al. (Heo et al., 2026)	full KV (ridge)	top- k layers, de-RoPE keys	no	retention conflates mechanics and mapping
KVComm (Shi et al., 2025)	selected KV layers	attention ranking	no	ranked layers carry the signal
Interlat (Du et al., 2025)	last hidden state	latent message	no	the last layer summarizes the cache
HCache (Gao et al., 2025)	hidden states	translator	no	states determine caches
This paper	full cache (zero re-prefill)	seven families incl. RAT	no	<i>identity-controlled evaluation</i> + diagnostic probes

2.4 FROZEN-MODEL CONTROLLABILITY

Task vectors compress in-context learning into a single mid-layer activation (Hendel et al., 2023); gist tokens compress prompts into a few virtual tokens (Mu et al., 2023); the tuned lens reads intermediate layers through affine probes (Belrose et al., 2023); and the FFN-as-memory view assigns factual capacity to feed-forward weights (Geva et al., 2021). Frozen students can therefore absorb injected information, which makes the H3 null result informative rather than tautological: the failure is not that frozen students cannot be steered, but that, in our tested teacher–student regime, raw KV translation does not deliver a usable steering signal to the frozen student. More transferable control signals may be easier to access in residual-stream directions or task-vector representations than in per-head KV entries.

2.5 WHAT THE SURVEY DOES NOT ESTABLISH

Across the four lines, no work separates mechanics from mapping from exploitability, injects the student’s own cache as an identity control, reports accuracy at matched perplexity, or probes exploitability without a translator. The claims of the translation line rest on pipelines whose components have never been isolated; the audit below supplies the missing instruments.

3 THE THREE-HYPOTHESIS AUDIT FRAMEWORK

3.1 SETUP

Let a teacher T prefill a context x and form a cache $C_T(x) = \{K_\ell^T, V_\ell^T\}_{\ell=1}^{L_T}$ with L_T layers, H KV heads, and head dimension d . A translator g_θ maps it to $\widehat{C}_S(x)$ in the space of a student S with L_S layers. The student then answers a query q with only the query tokens as input, $\hat{y} = S(q \mid \widehat{C}_S(x))$, under strict zero re-prefill: the context never enters the student’s forward pass. Three references anchor the audit: *student self-prefill* $S(q \mid C_S(x))$ is the baseline the handoff must not degrade; *identity injection* feeds $C_S(x)$ itself through the pipeline, isolating mechanics from mapping; *teacher full* $T(q \mid C_T(x))$ is the capability upper bound. We measure answer quality by the gold-letter probability p_{gold} , accuracy, and CHG = $p_{\text{gold}}(\text{handoff}) - p_{\text{gold}}(\text{student self})$, with 95% bootstrap intervals and sign-flip permutation tests; perplexity over the scoring suffix is recorded but never accepted as evidence of capability (Section 5.5).

3.2 HYPOTHESES, INSTRUMENTS, GATES

Figure 1 states the hypotheses and gates. **H1**: the identity injection matches the student’s own prefill per sample (gate: mean logit cosine = 1 within float tolerance). **H2**: some translator g_θ in the audited family $\mathcal{G}$ reaches replacement level (gate: the CI lower bound of CHG is above a small tolerance $-\epsilon$; a stricter *gain* gate additionally requires the CI lower bound > 0). Replacement asks only that the student is not degraded; gain asks for a measurable lift. We fix $\epsilon = 0.02$ gold probability before evaluation—about 4% relative to the 0.503 student baseline, a practical serving tolerance—making the replacement gate a standard non-inferiority test of $H_0: \text{CHG} \leq -\epsilon$ against $H_1: \text{CHG} > -\epsilon$; unlike a naive significance test, it does not additionally require the point estimate to be non-negative. **H3**: the teacher’s cache contains student-readable advantage (gate: some oracle probe configuration with teacher content improves over the student’s own cache). H3’s probe does not depend on training a translator: it directly measures what a frozen student can extract from teacher-cache content under the most favorable mixing.

4 AUDIT METHODOLOGY

4.1 IDENTITY CONTROL

At evaluation time we capture the student’s own cache over the context (allowed in calibration and probe stages) and feed it through the same mapper interface, cache construction, position handling, and scoring path as a translated cache. Any deviation from the student’s own prefill is mechanics loss—the control that prior evaluations omit.

4.2 DISJOINT SPLITS, UNIFIED PROMPTS, AND HONEST COUNTERS

Mapper calibration and evaluation draws are disjoint samples of the same distribution (HellaSwag and ARC-Challenge, shuffled then split); all methods share one token-level prompt format, so the only difference between the handoff and the student baseline is where the context representation comes from. Zero-prefill counters read the actual cache length before and after the scoring forward pass, and the scoring pass uses an isolated cache because the deployed transformers version mutates a cache even under `use_cache=False`, silently polluting co-located measurements.

4.3 MAPPER LADDER

We audit seven families under one interface. **Ridge (per-head)**: a 128×128 ridge map per student layer and head with Gram aggregation over variable-length calibration samples. **Affine (per-head)**: the same with mean centering and an intercept, fit in closed form over concatenated calibration samples. **Affine (per-layer)**: heads merged into rows, eight times fewer parameters. **Task-aware**: ridge initialization plus a bounded task fine-tune that mixes reconstruction with gold-answer cross-entropy (weight $\beta=0.3$) under a diagonal parameterization with a per-layer bounded gate. **RAT (residual-anchored translator)**: an architecture-derived map built from the two models’ own value projections, $W^{(s,h)} = [\pi(W_V^{S,h}) R \pi(W_V^{T,h})]^\top$ with $R = \arg \min_R \|E_T R - E_S\|_F^2 + \lambda \|R\|_F^2$, where $\pi(\cdot)$ is the pseudo-inverse pullback from a head’s value space to the residual stream and R aligns the residual streams through the shared 151,936-token embedding matrices (which both Qwen3 models ship). A closed-form rank- r correction fits the calibration residual, a Hungarian matching on head covariance profiles replaces the identity head correspondence, and the position-0 sink, whose translated value is a PPL catastrophe (Section 5.3), is overwritten with student statistics. **Joint MLP (over-parameterized)**: for each student layer, a two-hidden-layer MLP with width 256 maps the concatenation of all aligned teacher layers and all KV heads at a position to all student heads, giving about 37M parameters across K and V, five times the per-head affine family; it is trained on the calibration split by Adam for 300 epochs. Keys follow the same construction on W_K ; every family is evaluated at calibration budgets 30 and 200 with disjoint evaluation samples ($n=30$ and $n=63$ respectively).

4.4 ORACLE PROBES

These probes are diagnostic interventions rather than universal upper bounds: they give the student access to its *own* cache and introduce translated teacher content under controlled fractions and layer windows. A positive result would establish exploitability under that intervention; a negative result constrains—but does not rule out—nonlinear or jointly structured extraction paths (Section 8).

Fraction probe: $\alpha\hat{C}_S + (1 - \alpha)C_S$ for $\alpha \in \{0.25, 0.5, 0.75\}$ with $\alpha=0$ the student’s own cache.

Window probe: translated content in only the bottom, middle, or top third of student layers, student content elsewhere. Because every configuration keeps the student’s own cache dominant or localized, any improvement over $\alpha=0$ would indicate exploitable teacher-cache content; monotone degradation under the tested translators indicates no exploitable content within this mixture family; paths outside it are not covered (Section 8). Probes require the student’s own prefill, so they are flagged non-deployable and exist only as measurement instruments.

4.5 PAIRS, DATA, AND PROTOCOL DISCIPLINE

Primary pair: Qwen3-4B (36 layers, hidden 2560) $\rightarrow$ Qwen3-1.7B (28 layers, hidden 2048); contrast pair: $\rightarrow$ Qwen3-0.6B (hidden 1024); scale-extended pair: Qwen3-8B (36 layers, hidden 4096) $\rightarrow$ 1.7B and 0.6B. All share the same tokenizer, vocabulary, $H=8$ KV heads, $d=128$, and RoPE base 10^6 . Every run records its git commit and protocol version; every number below is read directly from a recorded run. We use multiple-choice QA deliberately: it gives a gold-probability control (the correct option’s log-probability is comparable across cache states) and makes disjoint calibration/evaluation splits enforceable without leakage. We additionally check a long-context needle MCQA (Section 5).

5 RESULTS

5.1 H1 PASSES: MECHANICS IS NOT THE FAILURE MODE

The identity control reproduces the student’s own prefill exactly: per-sample logit cosine 1.000, gold CHG +0.0001 $[-0.0006, +0.0009]$, $n=30$. The injection pipeline, cache construction, GQA head handling, and position accounting are therefore correct, and every subsequent gap is attributable to the translation itself. The same control verified the disk-persistence path, which reproduces the offline run bit-for-bit.

The identity control also fixes the perplexity protocol used below. Injected caches are scored on the suffix conditioned on the cache, so the identity-injection PPL (21.2) is the student’s own suffix PPL and matches the native suffix distribution by construction. The student self-prefill row of Table 2 reports a different quantity—the native loss averaged over the full context and suffix in one pass—so its value (25.4) is not expected to equal the injected-cache PPL. The H1 identity criterion is defined on the answer-scoring logits, where the two paths agree to float tolerance.

5.2 H2 FAILS FOR THE STRONG STUDENT; THE WEAK-STUDENT RESULT IS INCONCLUSIVE

Figure 2 and Table 2 give the landscape. Centering is the single biggest gain: per-head ridge (-0.296) drops to per-head affine (-0.138), roughly halving the deficit. Teacher and student value distributions differ in location and scale before they differ in content (norm ratios $1.58\times$ for keys, $7.44\times$ for values). The calibration budget, by contrast, buys nothing. Holding the evaluation set fixed ($n=100$) and sweeping the calibration budget, affine mapping keeps CHG inside $[-0.257, -0.238]$: -0.251 at $c=10$, -0.249 at $c=30$, -0.238 at $c=60$, -0.257 at $c=100$, and -0.244 for the full calibration pool (budgets beyond the pool size saturate and reproduce the full-pool result exactly), a spread of 0.019; this full-pool CHG of -0.244 is measured on the fixed evaluation set ($n=100$), while Table 2 reports -0.239 for the same calibration budget under the original disjoint-split protocol ($n=63$): two independent protocol reproductions, not one number written twice. The affine mapper appears calibration-saturated: within this function class, additional calibration data does not close the gap. Refitting the headline mapper on five independent calibration draws moves gold CHG within $[-0.292, -0.249]$ ($SD \approx 0.018$): the verdict is not an artifact of one calibration draw. Structure and function class change little either: per-layer, task-aware, RAT, and MLP mappers all sit inside the same band (Table 2). The per-head MLP is gradient-trained and seed-dependent: across independent

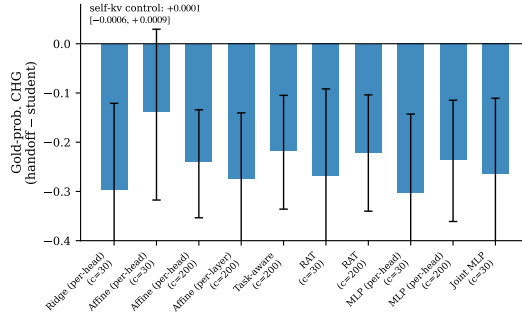


Figure 2: The mapper landscape for Qwen3-4B→1.7B under the unified audit protocol: gold-probability CHG over the student’s own prefill, with 95% bootstrap CIs. No family reaches zero; calibration budget moves affine mapping by only 0.005 on a fixed evaluation set; the architecture-anchored RAT, the nonlinear MLP, and the over-parameterized joint MLP sit inside the same band.

Table 2: Main audit results. Acc/gold/PPL: answer accuracy, gold-letter probability, and perplexity of the scoring suffix for the translated cache (kv-both). PPL entries for translated and identity caches score only the suffix conditioned on the injected cache; the student self-prefill row is the native loss over the full context and suffix in one pass and is not directly comparable (the identity-injection value is the student’s suffix PPL). CHG is the gold-probability change over the student’s own prefill with 95% bootstrap CIs. Pairs: Qwen3-4B→1.7B unless noted; 8B rows use the Qwen3-8B teacher; n_{eval} is the evaluation count (tail-protocol rows use a fixed set of 100). MLP rows are gradient-trained and seed-dependent: acc/gold/PPL are means over three independent trainings and the CHG column lists their range; all other rows are deterministic and reproduce bit-for-bit.

Configuration	Acc / Gold / PPL	Gold CHG [95% CI]	n_{eval}
Identity control (self-kv), 1.7B	0.500 / 0.503 / 21.2	+0.000 [−0.001, +0.001]	30
Student self-prefill, 1.7B	0.500 / 0.503 / 25.4	–	30
Teacher (full prefill), 1.7B	0.767 / 0.693 / –	–	30
Native average, 1.7B	0.167 / 0.169 / 536082.5	−0.334 [−0.508, −0.154]	30
Ridge per-head, c=30, 1.7B	0.200 / 0.207 / 88.5	−0.296 [−0.479, −0.121]	30
Ridge per-head, c=500, 8B→1.7B	0.310 / 0.288 / 146.1	−0.224 [−0.320, −0.123]	100
Affine per-head, c=30, 1.7B	0.367 / 0.365 / 46.1	−0.138 [−0.317, +0.029]	30
Affine per-head, c=200, 1.7B	0.254 / 0.258 / 56.4	−0.239 [−0.354, −0.134]	63
Affine per-layer, c=200, 1.7B	0.206 / 0.222 / 81441.1	−0.275 [−0.407, −0.140]	63
Task-aware, c=200, 1.7B	0.270 / 0.280 / 142.1	−0.216 [−0.336, −0.105]	63
RAT, c=30, 1.7B	0.233 / 0.235 / 501.6	−0.268 [−0.440, −0.092]	30
RAT, c=200, 1.7B	0.238 / 0.276 / 1008.2	−0.221 [−0.340, −0.104]	63
RAT no-anchor rank-32, c=30, 1.7B	0.267 / 0.280 / 23.7	−0.223 [−0.367, −0.088]	30
MLP per-head, c=30 (3 seeds), 1.7B	0.156 / 0.163 / 80.0	−0.39 to −0.26	30
MLP per-head, c=200 (3 seeds), 1.7B	0.254 / 0.258 / 122.7	−0.26 to −0.23	63
Joint MLP, c=30 (3 seeds), 1.7B	0.189 / 0.219 / 28.9	−0.30 to −0.26	30
Affine per-head, c=200, 4B→0.6B	0.302 / 0.283 / 71.8	+0.010 [−0.081, +0.102]	63
Affine per-head, c=30, 8B→0.6B	0.270 / 0.269 / 99.0	−0.023 [−0.104, +0.064]	100
RAT, c=200, 4B→0.6B	0.206 / 0.232 / 790.0	−0.041 [−0.120, +0.038]	63
Affine per-head, c=30, long-context	0.233 / 0.232 / 6554.1	−0.285 [−0.534, −0.025]	30

trainings its CHG spans −0.39 to −0.26 at $c=30$ and −0.26 to −0.23 at $c=200$, ever inside the same negative band, and it does not close the gap. An over-parameterized joint mapper (~37M parameters) that reads all teacher layers and all KV heads also fails: across three training seeds its CHG spans −0.26 to −0.30 with accuracy 0.189, even though it reconstructs a fluent cache (PPL 28.9 versus 25.4 for the student), reinforcing the perplexity–accuracy decoupling. The strong student never approaches its own prefill (0.503); the best mapper reaches 0.365. Scaling the teacher to 8B under the same protocol does not change the verdict: self-injection remains an identity (logit cosine 0.99998) and the ridge mapper reaches CHG −0.224 [−0.320, −0.123] at $c=500$, so H2’s failure is robust to teacher scale. The weak 0.6B student is qualitatively different. Affine at $c=200$ gives a near-parity point estimate +0.010 [−0.081, +0.102], and 8B→0.6B gives −0.023 [−0.104, +0.064];

Table 3: Cross-architecture robustness checks. Qwen3-4B teacher to non-Qwen and Qwen2.5 students under the within-family protocol (HellaSwag / ARC, $c=30$, $n=100$); identity injection passes in every row (self-kv CHG within ± 0.001). “KV h/d” gives the student’s KV heads and head dimension; the rectangular mapper mean-pools head groups and projects the head dimension. Rows marked “token-aligned” align teacher KV to student token positions by character overlap before fitting. No row recovers the teacher’s gold probability (0.691).

Student	KV h/d	Student acc / gold	Translated acc / gold	Gold CHG [95% CI]
Llama-3.2-1B	8/64	0.300 / 0.254	0.260 / 0.254	+0.000 [−0.012, +0.012]
Gemma-2-2B	4/256	0.270 / 0.257	0.220 / 0.248	−0.009 [−0.020, +0.002]
Llama-3.2-3B	8/128	0.390 / 0.281	0.200 / 0.237	−0.044 [−0.073, −0.013]
Qwen2.5-1.5B	2/128	0.400 / 0.385	0.200 / 0.228	−0.157 [−0.209, −0.106]
Gemma-3-1B	1/256	0.360 / 0.318	0.270 / 0.257	−0.061 [−0.135, +0.008]
Llama-3.2-3B (token-aligned)	8/128	0.390 / 0.281	0.280 / 0.257	−0.024 [−0.047, −0.004]
Gemma-3-1B (token-aligned)	1/256	0.360 / 0.318	0.250 / 0.245	−0.073 [−0.134, −0.021]

both intervals include zero, but both lower bounds fall below the pre-specified replacement margin $-\epsilon = -0.02$, so non-inferiority is not established at these sample sizes.

The failure is not specific to short commonsense contexts. On a long-context needle-in-a-haystack multiple-choice task (contexts of about 1024 tokens, $n=30$), where the teacher answers from a retrievable fact (0.733 accuracy versus the student’s 0.533), affine translation again fails: accuracy drops to 0.233 with gold CHG $-0.285 [-0.534, -0.025]$, while identity injection remains exact (CHG $+0.000 [-0.001, +0.001]$). The deficit therefore tracks the translation regime rather than the context length or task format.

Cross-architecture checks relax the one-family scope (Table 3). Replacing the Qwen3 student with five models from other families—Llama-3.2-1B and 3B, Gemma-2-2B, Gemma-3-1B, and Qwen2.5-1.5B, spanning one to eight KV heads and head dimensions 64 to 256—mechanics still hold (self-kv CHG within ± 0.001) and translation never recovers the teacher’s 0.691 gold. Four students are harmed (CHG -0.009 to -0.157); the weakest, near-chance student (Llama-3.2-1B) is non-inferior but gains nothing. Head groups are mean-pooled and head dimensions projected by the rectangular mapper; because the models tokenize independently, calibration pairs are matched position-by-position, so we read these as mechanism and replacement checks rather than fine-grained capability measurements. To address the tokenization confound directly, a token-aligned variant maps each student token to the teacher token with maximum character overlap before fitting and gives the same verdict on genuine cross-tokenizer pairs (Llama-3.2-3B: CHG $-0.024 [-0.047, -0.004]$; Gemma-3-1B: $-0.073 [-0.134, -0.021]$), so the negative result is not an artifact of independent tokenization; Qwen2.5 shares the Qwen tokenizer, making alignment a no-op there.

5.3 RAT ABLATIONS SEPARATE FLUENCY FROM CAPABILITY

RAT ablations separate fluency from capability: removing the analytic core repairs perplexity (PPL 23.7, within 2.5 of identity) while accuracy stays at 0.267—fluency recovery does not touch the answer deficit (Appendix C). RAT’s role is to rule out the objection that the failure is an artifact of insufficiently architecture-aware translation.

5.4 H3: NO TEACHER-CONTENT PROBE IMPROVES ON THE STUDENT

The probes answer H3 without training a translator (Figure 3). With 75% student cache and 25% teacher content, gold probability falls from 0.5031 to 0.3798, monotonically through $\alpha=0.75$; window injection localizes the damage—top-third replacement is indistinguishable from the student (0.5022 versus 0.5031), while the middle and bottom thirds are strongly harmful (0.2653, 0.2242). This is a sensitivity profile: the student’s behavior is substantially more sensitive to teacher-cache intervention in early and middle layers than in the top third. It does not by itself locate where the answer is routed; early and middle layers may also be more sensitive to off-manifold perturbations. An eight-way octant sweep ($n=30$) localizes the sensitive band: among the eight eighths of student layers, only layers 10–13 have a gold-probability interval excluding zero (CHG $-0.186 [-0.322, -0.060]$, gold 0.317 versus 0.503), while the top three eighths are statistically indistinguishable from the

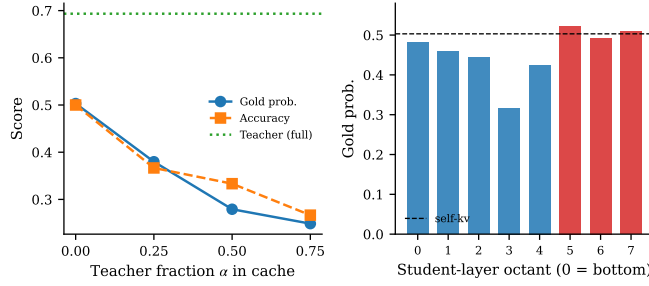


Figure 3: Oracle probes on Qwen3-4B→1.7B. Left: blending teacher-cache content into the student’s own cache (RAT translator) degrades gold probability and accuracy monotonically in the teacher fraction α ; $\alpha=0$ is the student’s own cache. Right: per-octant window injection (affine translator; octant 0 = bottom eighth of student layers); only layers 10–13 (octant 3) are clearly harmful, while the top three octants are indistinguishable from the student.

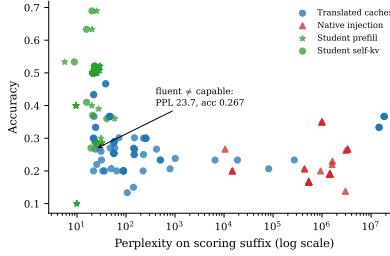


Figure 4: Perplexity over the scoring suffix versus accuracy across all methods and runs (log scale). The identity control and the student sit at low PPL and high accuracy; translated caches span two regimes; the best-reconstructed cache (PPL 23.7) remains 23 accuracy points below the student.

student (CHG -0.011 to $+0.020$). The probes were run with two translators—RAT and affine (the best in the ladder, gold 0.365)—with the same monotone decline: affine gives -0.086 (CI $[-0.18, +0.00]$) at $\alpha=0.25$ and -0.211 (CI $[-0.41, -0.02]$) at $\alpha=0.75$. The verdict is robust to translator quality: even the best mapper produces content the frozen student cannot exploit.

Direct native-cache injection provides an additional negative control: injecting the teacher’s native (norm-rescaled) cache entries degrades the student to -0.334 $[-0.508, -0.154]$ ($n=30$), replicated under the fixed-evaluation protocol (-0.266 $[-0.363, -0.168]$ at $c=30$; -0.251 $[-0.351, -0.152]$ at $c=200$); even untranslated teacher content in student coordinates does not help.

5.5 PERPLEXITY AND ACCURACY DECOUPLE

Figure 4 plots perplexity against accuracy for every run: the identity control and the student occupy the top-left regime, translated caches populate a broad band—including near-native perplexity with a 23-point accuracy deficit—and native injection sits at extreme perplexity (5.4×10^5) with low accuracy. Perplexity measures whether the cache permits fluent continuation; accuracy measures whether it supports the task. The two dissociate, so perplexity cannot serve as evidence of translation quality—a practice that appears in the serving literature because perplexity is cheap.

6 WHY TRANSLATION FAILS: AN ARCHITECTURAL ANALYSIS

We read this section as an architectural explanation, not an impossibility result: the per-head argument does not cover exploit paths that read the full multi-head, multi-layer cache (Section 8).

A cache entry is a linear projection $W_V h$ of a residual-stream state h ; translation must recover h from V , move it across the residual-space gap (2560 to 2048), and re-project. Two facts bound the recovery. First, the pullback $\pi(W_V)V$ returns only the row-space component of h ; the orthogonal

component—which can dominate a 2560-dimensional state observed through a 128-dimensional head—is invisible in that head’s value entry, so the per-head student value is underdetermined by its teacher counterpart: the mapping task is intrinsically underdetermined, and a learned translator can at best track the conditional mean. The mapper-fitting view quantifies this: held-out reconstruction by the fitted per-head affine mapper recovers student key states (mean $R^2 = +0.81$) but only partially recovers value states (mean $R^2 = +0.32$): roughly two-thirds of the value variance in the student’s per-head cache is not linearly recoverable from the teacher’s. The unit is a held-out calibration context: we fit on 20 contexts and score 10 disjoint held-out contexts; each yields a per-(layer,head) R^2 between predicted and true student states (pooled over token positions), and we report the mean over heads, layers, and the 10 contexts. The repository’s per-head ridge baseline agrees on the same held-out contexts (pooled $R^2 = +0.92$ for keys, $+0.29$ for values). Second, the teacher–student capability gap is plausibly dominated by parameter-mediated computation that raw KV transfer does not reproduce. FFN layers carry the model’s factual associations (Geva et al., 2021), and the Qwen3-4B→1.7B capability gap is a gap in these parameters, which a cache does not transport.

7 IMPLICATIONS FOR CACHE-TRANSLATION RESEARCH

Any future cache-translation evaluation should ship three instruments:

- an *identity control* that injects the student’s own cache through the full pipeline and must reproduce the student’s own prefill;
- a *gold-probability metric* with bootstrap intervals and permutation tests, reported as the change over the student’s own prefill;
- *oracle probes* that blend teacher-cache content into the student cache at controlled fractions and layer windows, probing whether teacher content is exploitable without training a translator.

Two gates apply: a replacement gate (CI lower bound above the pre-specified tolerance $-\epsilon$; the point estimate need not be non-negative) and a stricter gain gate (CI lower bound above zero). A pipeline delta over a student baseline is not evidence of translation, and a perplexity gain is not evidence of capability. Each instrument costs one evaluation run and composes with any serving stack.

For deployment, the weak student presents a near-parity regime rather than a demonstrated capability transfer. At 0.6B, translated injection is statistically indistinguishable from self-prefill (4B→0.6B: $+0.010 [-0.081, +0.102]$; 8B→0.6B: $-0.023 [-0.104, +0.064]$), but neither interval clears the pre-specified non-inferiority margin, so replacement-level service is not established; absolute accuracy is also near chance (0.302 versus 0.25 random), so this is not an endorsement. Top-layer injection is harmless—and buys nothing—so a system that must spend translation capacity can spend it there. Whether a capacity boundary separates a near-parity regime from failure is the question the 8B and 0.6B pairs begin to answer. For strong students, cache translation is currently a net negative; the teacher’s summary of the context scores 0.367 (gold 0.372) against the student’s 0.500 (gold 0.503) baseline,¹ so compression loses less than translation does.

The survivable versions of the idea move work off the cache and onto the weights or the replay: allow a target-side correction pass the way MoT does, train student-side adapters, or translate through a shared latent space anchored by task vectors. Each option gives up part of the cold-start guarantee and deserves a fresh identity control before its delta gets attributed to translation.

8 LIMITATIONS

Our audit is centered on one family (Qwen3) with four scale pairs, plus a cross-family check (five students, Table 3) and the HellaSwag / ARC-Challenge and long-context needle tasks, with $n=30$ –100 samples per run; the CI widths reflect that. The cross-family check matches positions after independent tokenization and its student is near chance, so it establishes mechanism and replacement, not capability. The oracle probes constrain exploitability within linear blending and three-layer

¹Text-channel numbers come from the v12-era in-run ablation ($n=30$); per-sample scores were not persisted for that run, so no bootstrap CI is available.

windows; a nonlinear exploit path outside this family would not be detected, though it would also need to survive the same measurement protocol. RAT’s analytic core assumes value projections are informative pullbacks of the residual stream; the real-model failure is itself an audit finding. We did not re-implement MoT’s full replay pipeline; our claims concern the zero-re-prefill regime. H2 is scoped to translators trainable from the disjoint calibration budget we test (≤ 200 examples), which includes the common linear and nonlinear families and a 37M-parameter joint MLP. The mapper-ladder results use a single seed per configuration; because the mapper fit is deterministic given its calibration draw, five independent calibration draws bound mapper-training variance directly: gold CHG stays within $[-0.292, -0.249]$ ($SD \approx 0.018$), inside the per-run bootstrap intervals, and identical configurations reproduce results bit-for-bit. The gradient-trained per-head MLP is the exception: its value is training-seed dependent (range in Table 2), so we report it as a range rather than a single number; every deterministic family reproduces exactly.

9 CONCLUSION

Cache translation promises to move a teacher’s work into a smaller model; the audit separates that promise into three testable parts. The mechanism holds up: an identity control reproduces the student’s own prefill exactly. The tested maps do not reach replacement: seven mapper families, a calibration ladder, and an architecture-anchored translator leave the strong student below its own prefill, while the weak student’s near-parity point estimate remains inconclusive under the pre-specified margin. The map, moreover, is not where the problem sits: no probe configuration delivers a student-readable advantage, and near-native perplexity still fails the task. The evidence is consistent with a weight-mediated advantage: recovering the teacher’s edge from raw KV states exceeds the tested translators (Section 8); the framework lets the next claim be checked in a single run.

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A REPRODUCIBILITY

All numbers come from the `apcs` repository’s recorded runs (`reports/runs/*/inject-eval/`), each carrying its git commit, protocol version, configuration, and per-sample scores. An independent reproduction of the deterministic configurations—ridge, affine, per-layer, task-aware, RAT, joint MLP, the calibration ladder, probes, octant windows, and the cross-architecture checks—matches the recorded values within 0.002; the gradient-trained per-head MLP is seed-dependent and reported as a range (Table 2). The held-out reconstruction R^2 is produced by `scripts/measure_reconstruction_r2.py` (output under `reports/reconstruction_r2/`). The audit protocol is implemented in `apcs.inference.evaluator` with the flags `calib_eval_split`, `calib_shuffle`, `probe_mode`, and `native_rescale`; mapper families live in `apcs.mapper` (`math`, `joint`, `task_aware`, `rat`, the per-head MLP in `mlp`, the joint MLP in `joint_mlp`, and the cross-family rectangular map in `rect`); the long-context needle task loader is `needle_mcqa` in `apcs.data.hf_dataset`; persistence and the cold-start online phase in `apcs.inference.kv_store`. Unit tests cover the audit counters, the store round-trip, and the mapper interface.

B ADDITIONAL MEASUREMENTS

The teacher/student KV norm ratios are $1.58\times$ (keys) and $7.44\times$ (values), measured over 30 calibration contexts per layer. Norm rescaling of the native cache repairs neither its perplexity (5.4×10^5) nor its accuracy, ruling out scale as the primary failure. The unified prompt renders choices once; the earlier protocol rendered them twice for the student baseline, which inflated the confidence-based deltas reported in our own preliminary sweeps and motivated the unified format in the audit.

C RAT COMPONENT ABLATIONS

RAT component ablations: removing the sink override explodes suffix perplexity to 2.7×10^5 —the translated position-0 entry, which carries outsized attention mass, is a catastrophe unless overwritten with the student’s own statistics. The min-norm pullback of the analytic core is harmful on real models; the cache can be made to look right to the language-modeling objective without becoming right for the task.

D CHANNEL ASYMMETRY

Channel ablations at $c=30$ rank value-only (0.367) above key-only (0.300) above both (0.200) on accuracy, while perplexity orders them in reverse (88 versus 253 versus 1.8×10^7). Mapping keys repairs fluency and damages routing: the mapped keys steer attention to plausible but task-irrelevant positions. One hypothesis consistent with the asymmetry: mapped keys steer attention routing while mapped values carry content, so key errors misroute a fluent read while value errors corrupt it audibly.

E TARGET-SIDE REPLAY DIAGNOSTIC

The audit holds strict zero re-prefill fixed, while MoT’s reported quality additionally includes a target-side replay with a Context Correction Loss. To probe how much a replay pass alone contributes, we add a diagnostic that re-reads the context through the student on top of the translated cache before scoring; it is explicitly non-deployable because it violates zero re-prefill. On Qwen3-4B→1.7B (affine $c=30$, $n=100$), replay leaves accuracy at 0.290 and gold CHG at -0.232 [-0.333 , -0.136], statistically indistinguishable from the no-replay result (-0.249) and far below the student’s own prefill (0.520, CHG $+0.000$). A naive replay pass therefore does not recover the strong student; recovery requires either discarding the translated cache (re-prefill) or the trained correction MoT uses. We do not implement MoT’s Context Correction Loss, so we cannot attribute its gains to replay alone.